Exp no:9 Implement Dielectric free space boundary condition in Electric Field and find the relative permittivity of the different medium.

```
Enter the value of theta1=30

Enter the value of theta2=60

Enter the value of er1=7

  "THE VALUE OF ER2 IS"
   21.000000

-->
```

File Edit Format Options Window Execute ?

exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes

exp333.sci

```
clc;
clear;

theta_1=input("Enter the value of theta1=");
theta_2=input("Enter the value of theta2=");
medium_1=input("Enter the value of er1=");

tan_theta= tand(theta_1)/tand(theta_2);
medium_2= medium_1/tan_theta;
disp('THE VALUE OF ER2 IS',[medium_2]);
```

Calculation:

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$\theta_1 = 30^\circ$

$\theta_2 = 60^\circ$

$\varepsilon_{r_1} = 7$

The value of ER_2 is

$$\tan\theta = \frac{\tan(\theta_1)}{\tan(\theta_2)}$$

$$ER_2 = \frac{ER_1}{\tan\theta}$$

i) Compute tan ratio.

$$\tan\theta = \frac{\tan(30^\circ)}{\tan(60^\circ)} = \frac{0.577}{1.732} = 0.333$$

ii) Compute ER_2 :

$$ER_2 = \frac{7}{0.333} = 21.0 \approx 21.000000$$

Case 1: In this case, the dielectric medium has a relative permittivity (εr) of 2, meaning the dielectric reduces the electric field by half compared to the free space. This is typical for materials like mica or certain types of plastics. ● Free Space Electric Field (E1): 100 V/m ● Dielectric Electric Field (E2): 50 V/m

*exp 7.sci | exp6 case i.sci | exp6 caseii.sci | exp6 caseiii.sci | exp7 c1.sci

```
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theta_1=input("Enter the value of theta1=");
theta_2=input("Enter the value of theta2=");
medium_1=input("Enter the value of er1=");

tan_theta= tand(theta_1)/tand(theta_2);
medium_2= medium_1/tan_theta;
disp('THE VALUE OF ER2 IS',[medium_2]);

```
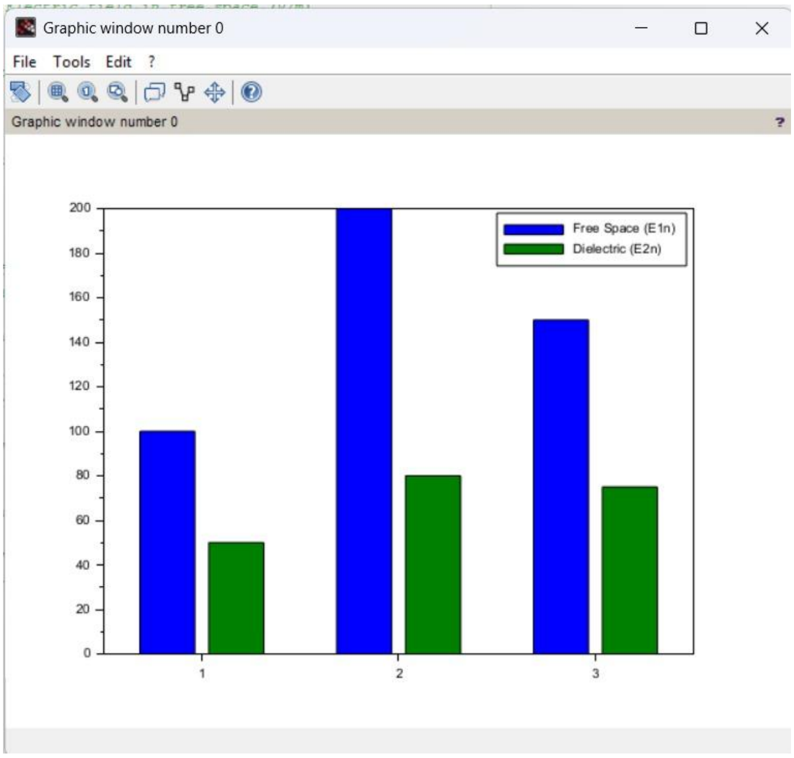

Output:

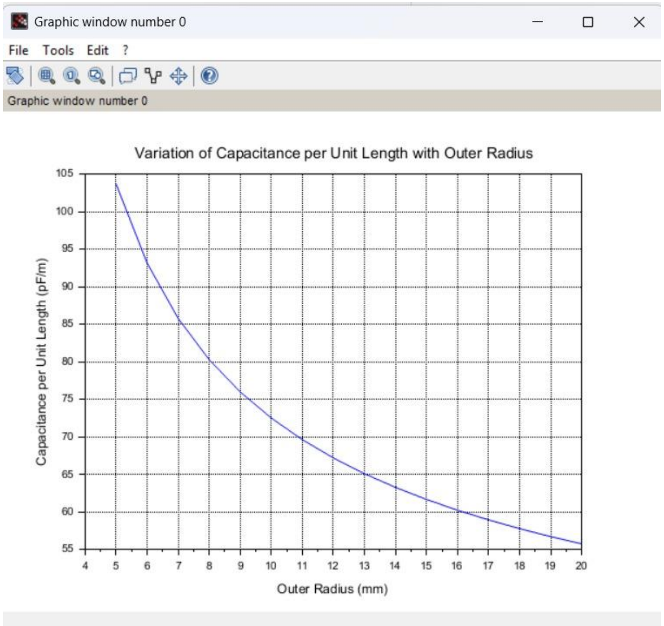


Case 2: Here, the dielectric medium has a higher relative permittivity (εr=2.5) compared to Case 1. This indicates a stronger polarization effect in the dielectric, as seen in materials like glass or water. ● Parameters: o En1=200 V/m (Electric field in free space). o E2n=80 V/m (Electric field in the dielectric medium).

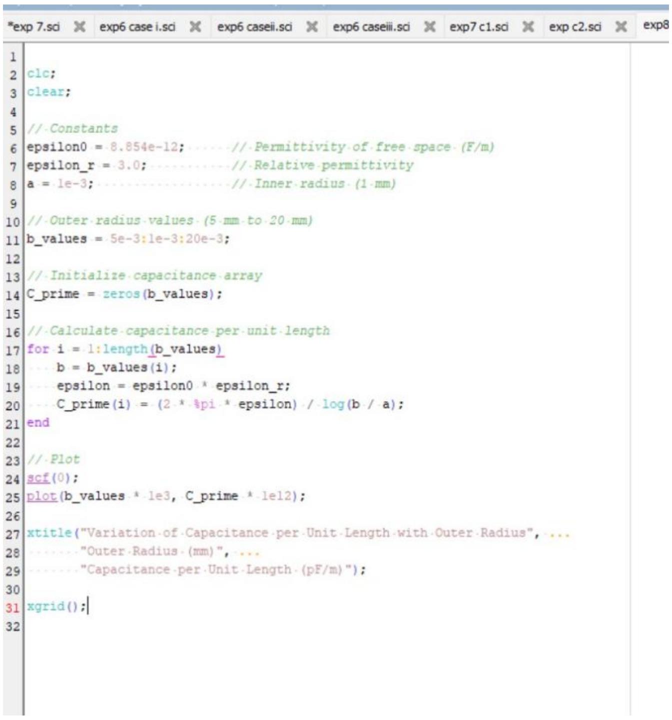
*exp 7.sci | exp6 case i.sci | exp6 caseii.sci | exp6 caseiii.sci | exp7 c1.sci | exp c2.sci | exp8

```

clc;
clear;

// Constants
epsilon0 = 8.854e-12;      // Permittivity of free space (F/m)
epsilon_r = 3.0;           // Relative permittivity
a = 1e-3;                  // Inner radius (1 mm)

// Outer radius values (5 mm to 20 mm)
b_values = 5e-3:1e-3:20e-3;

// Initialize capacitance array
C_prime = zeros(b_values);

// Calculate capacitance per unit length
for i = 1:length(b_values)
    b = b_values(i);
    epsilon = epsilon0 * epsilon_r;
    C_prime(i) = (2 * %pi * epsilon) / log(b / a);
end

// Plot
scf(0);
plot(b_values * 1e3, C_prime * 1e12);

xtitle("Variation of Capacitance per Unit Length with Outer Radius", ...
       "Outer Radius (mm)", ...
       "Capacitance per Unit Length (pF/m)");

xgrid();

```

Output:



Case 3: The dielectric medium here also has a relative permittivity of 2. The behavior of the material suggests it could be another low-polarization dielectric material, with a consistent reduction of the electric field by half. ● Free Space Electric Field (E1n): 150 V/m ● Dielectric Electric Field (E2n): 75 V/m.




```
clc;
clear;

// Constants
epsilon0 = 8.854e-12;       // Permittivity of free space (F/m)
epsilon_r = 3.0;            // Relative permittivity
a = 1e-3;                   // Inner radius (1 mm)

// Outer radius values (5 mm to 20 mm)
b_values = 5e-3:1e-3:20e-3;

// Initialize capacitance array
C_prime = zeros(b_values);

// Calculate capacitance per unit length
for i = 1:length(b_values)
    b = b_values(i);
    epsilon = epsilon0 * epsilon_r;
    C_prime(i) = (2 * %pi * epsilon) / log(b / a);
end

// Plot
scf(0);
plot(b_values * 1e3, C_prime * 1e12);

xtitle("Variation of Capacitance per Unit Length with Outer Radius", ...
       "Outer Radius (mm)", ...
       "Capacitance per Unit Length (pF/m)");

xgrid();
```

Output: